

SAP CPI Project 4 — Data Store Operations

Project Type: Persistent Message Storage & Retrieval

Status: Successfully Executed

Project Objective

This project demonstrates how SAP CPI Data Store operations work. The first integration flow stores XML order data into a CPI Data Store, while the second integration flow retrieves the stored message and logs it through Groovy scripting.

Core Concept

Until now, all CPI messages only existed during runtime execution. This project introduces persistent storage, where messages survive even after the integration flow has completed.

Integration Flow 1 — Write to Data Store

Flow Structure:

Timer Start → Create XML Payload → Write Data Store → End

Step 1 — Timer Start Event

The integration flow is triggered automatically using a Timer Start Event. A Timer was used because SAP CPI Trial environments may restrict external sender access.

Step 2 — Create XML Payload

A Content Modifier creates a sample XML order payload that simulates enterprise order data.

```
<Order>
  <OrderId>5001</OrderId>
  <Customer>Vyshnavi</Customer>
  <Country>India</Country>
  <Amount>2500</Amount>
</Order>
```

Step 3 — Write Data Store Operation

Configuration	Value
Data Store Name	write_data
Visibility	Global
Entry ID	5001
Overwrite Existing Message	Enabled

Expiration Period 30 Days

The XML payload is stored inside the CPI Data Store named "write_data" using Entry ID "5001".

Integration Flow 2 — Retrieve from Data Store

Flow Structure:

Timer Start → Get Data Store → Groovy Logging → End

Step 1 — Get Data Store Operation

Configuration	Value
Data Store Name	write_data
Visibility	Global
Entry ID	5001
Delete On Completion	Disabled

The second integration flow retrieves the previously stored XML payload using the same Data Store Name and Entry ID.

Step 2 — Groovy Logging

A Groovy script logs the retrieved message into SAP CPI Monitoring Attachments for debugging and monitoring purposes.

Retrieved Output

```
<Order>
  <OrderId>5001</OrderId>
  <Customer>Vyshnavi</Customer>
  <Country>India</Country>
  <Amount>2500</Amount>
</Order>
```

Important Concepts Learned

- Persistent Message Storage
- Data Store Write Operations
- Data Store Retrieval Operations
- Entry IDs and Message Lookup
- Global vs Integration Flow Visibility
- Stateful Integration Concepts
- Groovy Logging
- Monitoring Attachments

Issue Faced During Development

Initially, the retrieval flow failed because the Data Store Visibility was set to "Integration Flow". This restricted access only to the flow that created the Data Store entry.

The issue was resolved by changing Visibility to "Global", allowing multiple integration flows to access the same Data Store.

Real Enterprise Use Cases

- Retrying failed messages
- Temporary message buffering
- Caching API responses
- Asynchronous integrations
- Order persistence during system downtime
- Enterprise recovery mechanisms

Project Outcome

This project introduced stateful integration concepts in SAP CPI by enabling persistent storage and retrieval of integration messages. It also established the foundation for advanced enterprise integration patterns and prepares for upcoming Mapping and Transformation topics.